26 Monitoring and troubleshooting errors with logging

This chapter covers

- Understanding the components of a log message
- Writing logs to multiple output locations
- Controlling log verbosity in different environments using filtering
- Using structured logging to make logs searchable

Logging is one of those topics that seems unnecessary, right up until you desperately need it! There's nothing more frustrating than finding a problem that you can reproduce only in production and then discovering there are no logs to help you debug it.

Logging is the process of recording events or activities in an app, and it often involves writing a record to a console, a file, the Windows Event Log, or some other system. You can record anything in a log message, though there are generally two different types of messages:

- *Informational messages*—A standard event occurred: a user logged in, a product was placed in a shopping cart, or a new post was created on a blogging app.
- *Warnings and errors*—An error or unexpected condition occurred: a user had a negative total in the shopping cart, or an exception occurred.

Historically, a common problem with logging in larger applications was that each library and framework would generate logs in a slightly different